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(1) Research Program/Project/Study Title: Analyzing the Influence of Instructional Material in Technical English 2 on the Academic Achievement of Criminology Students	
Proponent Name:	Dr. Christa Hope G. Corros
Academic Rank:	Instructor 1
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:							
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								
(7) Sector/Commodity		(8) Discipline/Field of Specialization							
		Education							

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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

The significance of quality and comprehensive instructional materials within academia cannot be overstated, as they serve as indispensable tools for facilitating students' proximal development and aiding teachers in effectively scaffolding learning processes. At the tertiary level, where students are expected to engage in more independent learning, the presence of instructional materials assumes heightened importance, with teachers assuming the role of learning facilitators.

Vygotsky's sociocultural theory underscores the pivotal role of instructional materials in mediating learning experiences and fostering cognitive development. According to Vygotsky (1978), these materials serve as cultural tools that shape the learning environment and provide the necessary support for learners to achieve their potential within their zone of proximal development.

Also, research by Mayer (2001) emphasizes the cognitive theory of multimedia learning, which posits that well-designed instructional materials, incorporating visual and auditory elements, can enhance students' understanding and retention of complex concepts. Such materials not only cater to diverse learning styles but also stimulate deeper cognitive processing, thereby promoting more meaningful learning outcomes.

In alignment with these theoretical frameworks, the study conducted by Johnson and Smith (2018) on the effectiveness of instructional materials in tertiary education corroborates the assertion that comprehensive resources significantly contribute to students' academic success. Their findings underscored the positive correlation between the availability of quality materials and students' learning outcomes, highlighting the pivotal role of such resources in tertiary education settings.

This study builds upon the previous research conducted by the authors, titled 'Sufficiency and Utilization of Instructional Materials in the Teaching of Technical English 2 for Criminology Students in the Province of Capiz: Development of Comprehensive Instructional Material on Technical 2. The results on both the sufficiency and utilization of instructional materials in the teaching of Technical English 2 provide valuable insights into the current state of resources available to educators and how effectively these materials are integrated into the curriculum. Combining these results offers a comprehensive understanding of the challenges and opportunities in enhancing the teaching and learning experiences for criminology students. The results on sufficiency highlight those certain instructional forms, such as Affidavit of Arresting Officer, Affidavit of Witness, and Application/Petition to Wire Tapping, are lacking or have limited availability. These suggests a need for proactive efforts to develop and update instructional materials to address the gaps and ensure that students have access to comprehensive resources. (Corros and Caduco, 2023).

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project

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inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

Building upon the previous study's output, which involved the development of a Comprehensive Instructional Material for Technical English 2 tailored to Criminology Students to address the inadequacies of existing teaching materials, this study aims to evaluate the effectiveness of the aforementioned book and its impact on the academic performance of Criminology students.

Furthermore, this study seeks to refine and enhance the instructional material based on the findings, aiming to ensure its quality before publication and wider dissemination to schools nationwide offering Criminology programs.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study aims at determining the influence of instructional material in Technical English 2 on the academic achievement of criminology students.

Specifically, this study aimed to answer the following questions:

1. What is the academic performance of students in the pretests and posttests of the experimental and the control groups?
2. Is there a significant difference in the academic performance of students in the pretests and posttests of the control and experimental groups?
3. Is there a significant difference in the academic performance of students in the pretests and posttests of the control group?
4. Is there a significant difference in the academic performance of students in the pretests and posttests of the experimental group?
5. Is there a significant difference in the academic performance of students in the posttests of the control and experimental groups?

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of -the-art of current technology / information from which the project proposal will take-off).

According to Farombi, (1998), instructional materials include books, audio-visual, software and hardware of educational technology. He further opines that the availability, adequacy and relevance of instructional materials in classrooms can influence quality teaching, which can have positive effect on students' learning and academic performance. The insight from Farombi on linking instructional resources to students' academic performance serve critical in the provision of quality education (Tety, 2016).

In his study Adeogun (2001) revealed a strong positive link between instructional resources and academic performance. According to Adeogun, schools that possess more instructional resources performed better than schools that have less instructional resources. This finding supported the study by Babayomi (1999) that private schools performed better than public schools because of the availability and adequacy of teaching and learning resources. Adeogun (2001) noted that there was a low level of instructional resources available in public schools and hence commented that public schools had acute shortages of both teaching and learning resources. He further commented that effective teaching and learning cannot occur in the

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classroom environment if essential instructional resources are not available (cited by Tety, 2016).

Additionally, on the study conducted by Adalikwu & Iorkpilgh (2012) the research hypothesis showed that there is a statistical relationship between the academic performance of chemistry students and the use of instructional materials in Teaching-learning. The result agrees with the findings of Inyang (1997) that teaching is effective when the teacher make use of instructional materials. (Lance et al, 1999; Todd & Kuklthau, 2004.) Confirmed a significant correlation between the presence and the use of library materials by the students and teachers with better performance. Similarly, (Todd & Kuklthau, 2005, p.82.) found a simple correlation between the students' inputs and better academic achievement.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

This study is anchored on instructional model designs and theories. According to Kurt (2016), an instructional design model provides guidelines to organize appropriate pedagogical scenarios to achieve instructional goals. Instructional design can be defined as the practice of creating instructional experiences to help facilitate learning most effectively. Driscoll & Carliner (2005) states that "design is more than a process; that process, and resulting product, represent a framework of thinking" (p. 9).

Similar ideas are held by Lev Vygotsky, a Russian psychologist who held a view that tools and signs, which are in a form of instructional materials, have the capacity to develop in students' higher level of thinking, which is important in problem-solving activities. However, since they are considered to be domain-specific, the ways instructional materials can start cognitive development is yet to be studied with respect to classroom teaching (Tety, 2016).

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The present study will be conducted to compare the influence of Instructional Material in Technical English 2 on the Academic Achievement of Criminology Students of Filamer Christian University in the Academic Year 2023-2024.

This Quasi-Experimental research will use the pretest-posttest non-equivalent control group design. According to Best and Khan as cited by Tropico 2015, Quasi-experimental design resembled the true experiment design, however it does not employ random assignment of the participants either in the control or experimental group. This was commonly used in classroom experiment when the experimental and control groups are such naturally assembled group as intact classes that may have "collective similarity" in characteristics.

The diagram that follows presents the pretest and posttest non-equivalent control group design.

Time

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Experimental Group 01 ----- 03

Control Group 02 ----- 04

In the pretest-posttest control group design, the intervention or treatment will be introduced to the experimental group, but withheld from the control group. Before the introduction of the intervention, a survey observation, and testing will be done in both the experimental group (01) and control group (02). After the intervention to the experimental group, another observation of testing (posttest) will be administered to both groups (03) and (01), respectively. The pretest will determine whether the two groups have truly "collective similarity" at the start of the experiment. The two posttests (03) and (04), will be compared (Castillo, as cited by Tropico 2015).

The Instructional Material in Technical English 2 will be used for the experimental group. The independent variables are the use of Instructional Material in Technical English 2 and the usual materials used by the instructor in teaching while the dependent variable is the academic achievement of the students.

The descriptive statistics that will be used in the study are the mean, and standard deviation. The inferential statistics is t-test set at 5% level of significance.

This study will be conducted during the Second Semester of the Academic Year 2023-2024 utilizing two (2) sections of the Criminology Students at Filamer Christian University. The participants will be grouped into two (2) with equal number of students in the experimental group and control group.

The researcher will use lottery sampling in the selection of the group under the use of Comprehensive Instructional Material in Technical English 2 and the group that will be subjected to the conventional teaching. The first number drawn will correspond to the section assigned as the experimental group. The second number drawn will correspond to the section assigned as the control group.

The concern about "collective similarity" in the assignment of students in the two groups prior to the intervention will be addressed by matching their examination scores on the Midterm Examination.

The researcher will use a researcher-made test to measure the academic performance of criminology students. A 50 (fifty) item multiple choice test with four (4) options will be made. A Table of Specification will also be prepared before writing the test items to make sure that each topic for the coverage of this study will be properly presented in the test. Prior to the use of the instrument, the instrument will undergo content validation and consultation with the

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English teachers and the Coordinator/Heads of the Department. All suggestions and recommendations will be incorporated in the assessment tools.

After the validation, the 50 (fifty) item multiple choice test will be pilot tested to thirty (30) criminology students of Capiz State University, Dayao Satellite College who were not included in the study. The result of the test will undergo item analysis and reliability testing using the Statistical Package for Social Science (SPSS) software. The analysis will be the bases on which item will be retained, revised and discarded. According to Navaro (2012), items with 0.26 - 0.75 difficulty index and 0.46 - 1.0 discrimination index will be retained.

To interpret the academic performance of the Criminology Students, the scales shown below will be used.

Scale	Description
28.01 – 35	Excellent
21.01 – 28	Very Good
14. 01 – 21	Good
7.01 - 14	Poor
0.00 – 7	Very Poor

The researcher will employ three phases in the conduct of the study: the pre-experimental phase, the experimental phase and the post-experimental phase.

Pre-experimental phase. In this phase, the researcher will secure the following:

First, the researcher will identify and list down the competencies related to Technical English 2 as mandated by CHED. After that, corresponding topics will be prepared for both for the experimental and control group.

In teaching the lessons, for the experimental group, the use of Comprehensive Instructional Material for Technical English 2 designed by the researchers will be implemented while for the control group, conventional teaching will be used.

Second, the researcher will construct a Table of Specifications (TOS) considering the topics that will be included in the experimental period. Based on the table of specification made, the test will be constructed. In order to ascertain the validity of the instrument, the test will be submitted to the Quality Assurance Chair, Program Chairperson and Dean of the Criminal Justice Education.

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Also, the test will be pilot tested for item analysis to ensure a moderate difficulty and discrimination index of the test.

Third, the researcher will ask the permission from the Dean of the Criminology Department of Filamer Christian University to use the two groups of students as participants of the study.

Finally, the test will be administered to both control and experimental groups. The result of the pretest will be checked, recorded and tabulated for analysis using the appropriate statistical tools, however the test materials will be kept safely together with the test questions for it will be the same set of tests that will be administered again at the end of the investigation period.

Experimental Phase. During the experimentation, one group will be assigned to experimental group and will be taught under the use of the Comprehensive Instructional Material for Technical English 2, while the other group will be assigned in the control group and will be taught under conventional teaching under similar environmental conditions.

The format will be used to both experimental and control groups.

Post-experimental Phase. After the experimental phase, the following tasks will be accomplished by the researcher; the data on the pretest and posttest will be tallied and encoded in the Microsoft Excel application. Both the pretest and posttests results will be processed using Statistical Package for Social Science (SPSS) software. The differences between the pretest and posttest results will be determined by submitting the data to statistical tools to answer the statement of the problem and the hypothesis stated in this study. Both tabular and textual presentations will be used in data presentation and analysis. The data gathered from the study will be analyzed using the Statistical Package for Social Sciences (SPSS) software using the appropriate statistical procedures:

Mean. This will be employed to determine the academic performance of criminology students.

Standard Deviation (SD). This test will be used to determine the homogeneity and heterogeneity of the scores obtained by the students.

t - test for dependent samples. This test set at 5% alpha level of significance will be used to determine the differences between the academic performance of students of the control and experimental group in both pretest and the posttest.

t - test for independent samples. This test set at 5% alpha level of significance will be used to determine the differences between the of students of the control group in the pretest and posttest and the academic performance of the experimental group in the pretest and posttest.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use grant chart.

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Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Field work/Survey												
2. Analysis of data gathered from observation												
3. Lay-outing, Editing and Printing of 100 copies of coffee table book												

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

This study seeks to develop an enhanced and comprehensive instructional material tailored for Technical English 2 specifically designed for criminology students. The objective is contingent upon demonstrating that the utilization of this material leads to improved academic performance among students.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Our endeavor to enhance criminology education through the development of a refined and comprehensive instructional material for Technical English 2 entails a multifaceted approach to technology transfer. We envision a process that begins with the production and distribution of physical copies of the instructional material to educational institutions nationwide offering criminology programs. Concurrently, we plan to organize training workshops aimed at equipping educators with the necessary skills and knowledge to effectively utilize the instructional material in their classrooms. Additionally, we recognize the importance of digital resources in modern education and thus intend to develop online platforms housing digital versions of the instructional material, along with supplementary materials and interactive tools to enrich the learning experience. As implementation progresses, we are committed to providing ongoing monitoring and support to educators, ensuring they receive the guidance needed to successfully integrate the instructional material into their teaching practices. Moreover, we value feedback as a catalyst for improvement, and thus, we will establish mechanisms to gather input from educators and students, fostering a culture of continuous refinement and enhancement. Through this comprehensive technology transfer activity

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design, we aim to facilitate widespread adoption and seamless integration of the enhanced instructional material, ultimately enhancing the quality of criminology education and empowering students to achieve academic excellence.

(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

The results of the study will be of great help to the following:

Criminology Students. Through engaging with the refined instructional material tailored specifically for Technical English 2, students can expect to enhance their comprehension and mastery of technical language and concepts relevant to criminology. This resource offers structured content and interactive tools designed to facilitate deeper understanding and retention of course materials, ultimately leading to improved academic performance. Additionally, students develop essential communication skills essential for success in the field of criminology, empowering them to effectively convey complex technical information with clarity and precision.

Educators. Those who are tasked with teaching Technical English 2 in criminology programs will also experience positive outcomes from this study. The availability of a comprehensive instructional material streamlines lesson planning and resource development, enabling educators to deliver high-quality instruction that resonates with diverse learning styles. The structured content and pedagogical guidance provided by the material enhance teaching effectiveness, fostering an environment conducive to student learning and engagement.

Administrators. Within educational institutions offering criminology programs can benefit from the implementation of this study by supporting the delivery of quality education in Technical English 2. The improved academic performance of students in Technical English 2 reflects positively on the institution's reputation and standing within the academic community, further reinforcing its commitment to excellence in criminology education.

Law Enforcement Agencies. As criminology students graduate with enhanced language proficiency and communication skills, they become valuable assets to law enforcement agencies, facilitating collaboration and knowledge exchange in addressing complex challenges in criminology and law enforcement.

Future Researchers. Can build upon the findings of this study to conduct further investigations into pedagogical approaches, curriculum design, and student engagement within Technical English 1 courses. Through studying the experiences of educators and students using the instructional material, researchers can identify best practices, challenges, and areas for improvement, contributing to ongoing scholarship in criminology education.

(17) Personnel Requirement

Name	Role	Field of Specialization
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	(Program, Project, Study Leader)	
Ms. Christa Hope G. Corros	Study Leader	Education
Dr. Sebastian C. Caduco, Jr.	Project Member	Criminology

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (Php)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		Php 1,000.00
- Meal allowance during data gathering		Php 5,000.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		Php 2,000.00
- Printing		Php 2,000.00
C. CAPITAL OUTLAY		
Total		Php 10,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Adalikwu, S., & Iorkpilgh, R. (2012). The Impact of Instructional Materials on Chemistry Students' Academic Performance. International Journal of Science Education, 28(1), 67-82.

Adeogun, O. (2001). Instructional Resources and Academic Performance in Secondary Schools. Journal of Educational Studies, 25(3), 123-137.

Babayomi, A. B. (1999). Comparative Analysis of Academic Performance in Public and Private Schools. Educational Research Quarterly, 36(4), 289-302.

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Farombi, J. G. (1998). Teaching Effectiveness: The Role of Instructional Materials. University of Lagos Press.

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Lance, D., et al. (1999). Library Resources and Academic Performance: A Correlational Study. Journal of Academic Libraries, 42(3), 189-204.

Mayer, R. E. (2001). Multimedia learning. Cambridge University Press.

Tety, K. (2016). Impact of Instructional Materials on Students' Academic Performance. Journal of Educational Research and Development, 15(2), 45-58.

Todd, R., & Kukulthau, C. (2004). Impact of Library Materials on Student Achievement: A Meta-Analysis. Educational Research Review, 30(2), 65-78.

Todd, R., & Kukulthau, C. (2005). The Role of Library Resources in Academic Success. Journal of Library and Information Science, 38(1), 79-94.

Vygotsky, L. S. (1978). Mind in society: The development of higher psychological processes. Harvard University Press.

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

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Academic History

Doctorate	Colegio de la Purisima Concepcion 2019 – 2022, Doctor of Philosophy in Education Filamer Christian University 2022 up to present, Doctor of Education in English Language Teaching
Masters	Filamer Christian University 2017 - 2019, Master of Arts in Teaching English
Tertiary	Capiz State University, Main Campus, 2012 – 2016, Bachelor of Secondary Education Major in English

Professional Experience:

- **Instructor 1**
Capiz State University , Dayao Satellite College
2020 to date
- Teacher 1

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Congressman Ramon A. Arnaldo High School
2017 - 2020

Licenses and Certification:

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Academic History

Doctorate

Masters

Tertiary

Professional Experience:

Licenses and Certification:

- Registered Criminologist